

Performance Evaluation of 1kW Asynchronous and Synchronous Buck Converter-based Solar-powered Battery Charging System for Electric Vehicles

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Abstract—This paper presents the design and evaluates the system performance of one-kilowatt capacity asynchronous and synchronous buck converter based solar-powered charging systems for battery-driven electric vehicles. The dc motor-operated three-wheeler rickshaw was taken for testing the systems, where a battery bank containing four series-connected sub-colloid storage type batteries of each with a capacity of 12V, 120Ah has been used. PSIM simulation software has been used to evaluate the performances of these two types of battery charging systems. Hardware prototypes of these two types of charging systems have also been made and an experimental testbed comprising a 48V battery bank of 100Ah capacity with a charging current of 6A was performed. The experimental results have also been evaluated and compared.

Keywords—Battery charger, Synchronous Buck Converter, Asynchronous Buck Converter, Electric vehicle, PWM.

I. INTRODUCTION

The relation between the economic development and transportation system is firmly analogous, however, the alarming news is that the present transportation system is heavily dependent on oil consumption in developing and underdeveloped countries. By the year 2030, the share of oil consumption in the transportation sector will increase up to 55%. To lessen this huge dependency on oil and to make a carbon emission-free environment, Electric Vehicles (EVs) would be a practical solution. Since EVs run by rechargeable batteries and are getting charged from the grid, an adequate charging mechanism must be followed to ensure the proper power quality [1]. In the South Asian region, Bangladesh is a developing country where air pollution rising in a threatening position for public health [2]. The mortality rate for air pollution is 15,000 people per year in Bangladesh. Furthermore, nearly 7 million people suffering from asthma, a respiratory distress among which more than half are children [3]. It is also stated that up to 3,500 lives can be saved and up to 230 million cases of respiratory diseases can be avoided annually which is equivalent to US\$ 500 million savings if air pollution can be reduced [4]. Thus, to pacify the greenhouse effect, the electric transportation system is the most proposing solution [5]. Moreover, EVs could reduce 35% to 50% Nitric Oxide (NO) and Reactive Organic Gases (ROG), 45% to 65% of Petroleum and 30% to 45% of greenhouse gases [6]. Till now in Bangladesh, 7% of the total population is deprived of electricity [7]. Most of these un-electrifying places are located in rural areas. The most common transport in rural areas are the rickshaw, van, and bicycle. To provide an environmentally friendly, silent transport system for these rural un-electrified areas, the light-weight, energy-efficient and improved electric motor-operated pedal cycle-rickshaws established its lucrative position alternative to petrol and diesel-powered three-wheelers. Besides, it ensures not only a large-scale employment to the common people but also provides a

fatigue-free extra income to the rickshaw puller [8], [9]. However, the conventional way of the battery charging process required the electric grid-connections which is generated from fossil fuel that causes again the problem of global warming and climate change. Moreover, conventional charging from the electric grid involves a step-down transformer to lower the grid's line voltage which is then rectified by a full-bridge rectifier. The rectified signal is then passed through the L-C low pass filter. Since the diode bridge rectifier has non-linear characteristics, it is responsible for the poor sending-end power factor of approximately 0.65 to 0.70 and the use of transformer produces more harmonics injecting into the grid, which makes it an inefficient charger [10].

To overcome these limitations by avoiding to connect conventional heavy-weight and low-efficient battery charger to national grid, in this paper, a simpler and reliable battery charging system by using renewable energy sources and buck converters [11-12] are presented. The automatic optimization of the converters duty ratio promotes the competent operation of charging unit of the proposed systems.

II. PROPOSED SYSTEM

The block diagram of the proposed battery charging method is depicted in Fig.1. The input DC power from the solar panel will be fed to the SBC and ASBC for lowering the output voltage to battery charging level, where the required PWM signals will be generated via the control unit. The output of this SBC and ASBC converter will then be fed to the 48V battery bank. Since the 20 kHz, PWM signals are generated by the PIC 16F690 microcontroller, these proposed systems only need to sense the load (battery bank) voltage to ensure constant charging current.

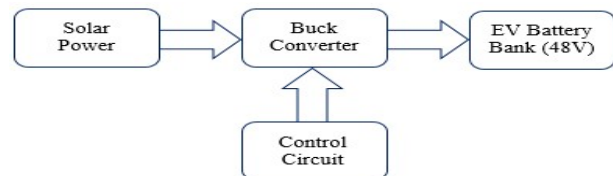


Fig. 1: Block diagram of the proposed charging system

The overall system design for the proposed battery charging system is described in the following subsections.

A. 1 kW Capacity Solar Module

To construct the 1 kW capacity buck converter (SBC and ASBC), the input power of the converter must also have to be 1 kW DC power. PSIM simulation software provides the solar module (physical model) utilities. This solar module follows the same configuration of the MSX-60 solar module from BP Solar. The designed parameter for 1 kW capacity solar module is shown in Table I.

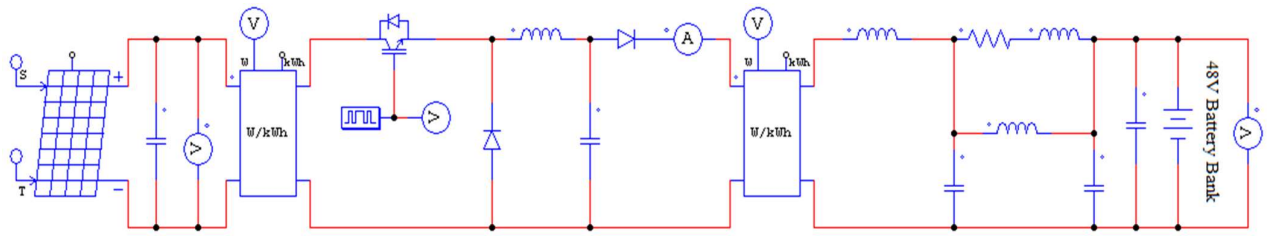


Fig. 2 PSIM simulation circuit of the proposed battery charging system using Asynchronous Buck Converter

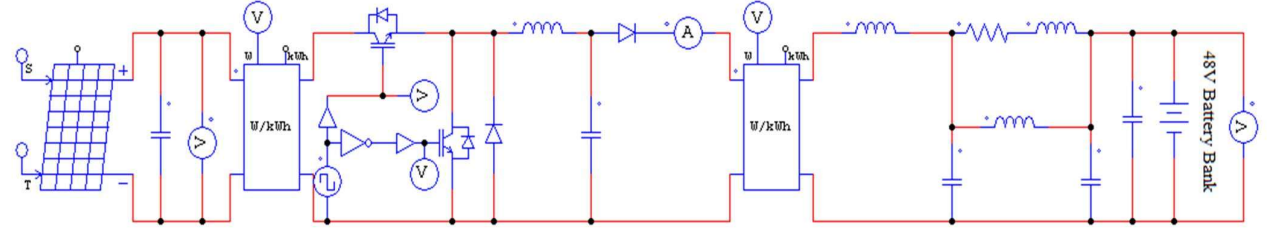


Fig. 3 PSIM simulation circuit of the proposed battery charging system using Synchronous Buck Converter

As each MSX-60 solar module comprises 36 cells, so 30 units of MSX-60 solar module were required in this project to make a 1 kW system.

TABLE I. Design Parameters of Solar Module for 1 kW solar system

Parameters	Value
Solar Cell Number	1080
Maximum Power	1816.05 W
Voltage at Maximum Power	511.14 V
Current at Maximum Power	3.55 A
Open-Circuit Voltage	21.1 V
Short-Circuit Current	3.8 A
Standard Light Intensity	1000 W/ m ²
Temperature Reference	25°C

B. Design of 1 kW Capacity Asynchronous and Synchronous Buck Converters

The design parameters of Asynchronous and Synchronous Buck Converters used in the proposed charging system are enlisted in Table II and III, respectively.

TABLE II. Design Parameters for Asynchronous Buck Converter

Parameters	Value
Input DC Voltage	543.50 V
Output DC Voltage	48 V
Maximum Output Current	26.04 A
Applied Switching Frequency	20 kHz
Filter Inductance	5 mH
Filter Capacitance	2000 μ F
Rated Capacity	1242.63 W

TABLE III. Design Parameters for Synchronous Buck Converter

Parameters	Value
Input DC Voltage	559.65 V
Output DC Voltage	48 V
Maximum Output Current	34.24 A
Applied Switching Frequency	20 kHz
Filter Inductance	5 mH
Filter Capacitance	2000 μ F
Rated Capacity	1623.75 W

III. SIMULATION RESULTS

The PSIM simulation circuits of the proposed battery charging system using Asynchronous and Synchronous Buck Converters using the design parameters listed in Tables I-III are depicted in Fig. 2 and 3, respectively.

Figure 4 shows the simulated output of input dc voltage (V input), gate signal (g1), load current and the output dc voltage level while charging

with the proposed Asynchronous Buck Converter. Here the switching frequency was considered as 20 kHz while the duty cycle for switching was considered as 8.3%. It also shows the Pulse Width Modulation (PWM) signal which is fed to the IGBT switch G1. Moreover, it shows the charging current along with the output dc voltage behave steadily.

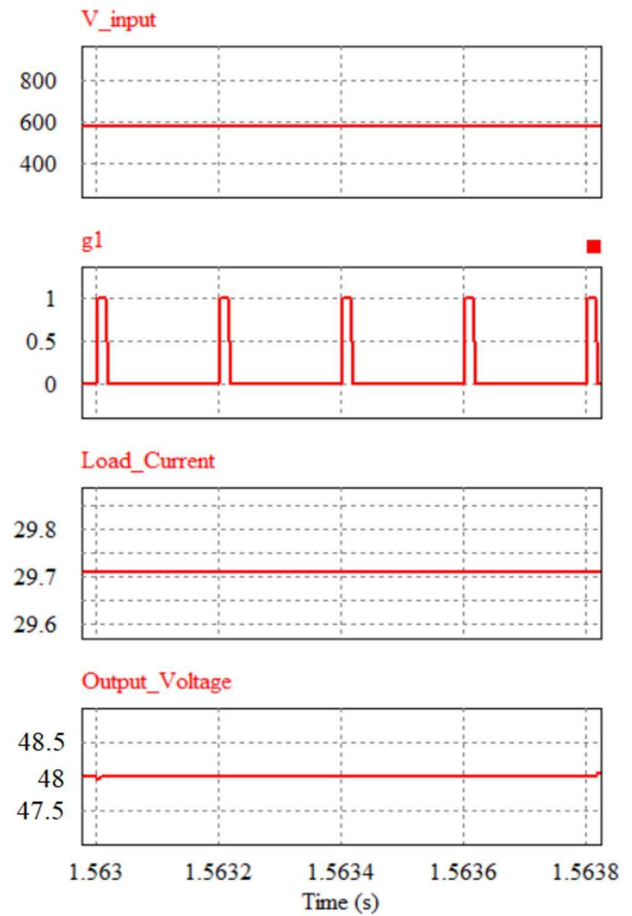


Fig. 4: PSIM simulated waveform showing the input dc voltage (V input), gate signal (g1), load current and the output dc voltage level while charging with the proposed Asynchronous Buck Converter

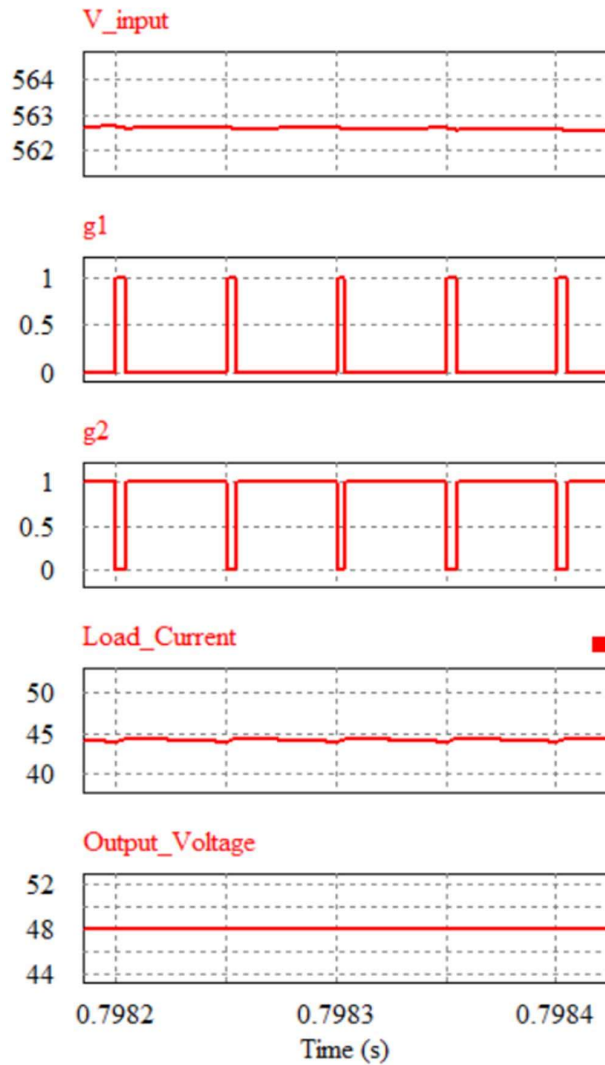


Fig. 5: PSIM simulated waveform showing the input dc voltage (V_{input}), gate signals ($g1$ and $g2$), load current and output dc voltage level while charging with the proposed Synchronous Buck Converter.

Figure 5 shows the simulated output of input dc voltage (V_{input}), gate signals ($g1$ and $g2$), load current and the output dc voltage level while charging with the proposed Synchronous Buck Converter. Here the switching frequency was considered as 20 kHz while the duty cycle for switching was considered as 8.6%. Moreover, it shows the charging current behaves steadily. However, the output dc voltage has some ripple which can be solved by using a small capacitor bank connected parallel to the battery bank (48V).

IV. HARDWARE PROTOTYPE

To experimentally verify the viability of these proposed charging systems, hardware prototypes of ASBC and SBC were developed as shown in Fig.6 and Fig.7, respectively. The overall experimental setup for charging the EV battery bank (48V, 100Ah) is shown in Fig.8.

A. Circuit Description

The PIC 16f690 microcontroller was used for generating PWM pulses, which were fed to Driver IR2104. The PIC microcontroller and Driver need some power to accomplish their work. This was done by 7805 and 7815 which provided

5V and 15V DC to PIC and Driver unit. Two alternative PWM signals coming out from the driver was fed to the IGBTs. Finally, after using the filter, the required DC output appears. For Asynchronous Buck Converter (Fig. 6) all the components are as same as Synchronous Buck Converter (Fig.7) except using single IGBT in the latter case.



Fig. 6: Prototype of proposed Asynchronous Buck Converter based charger

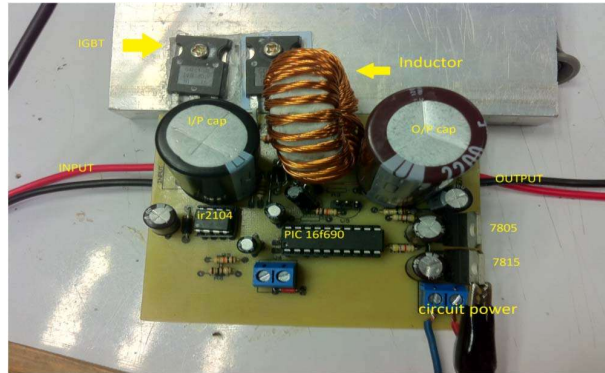


Fig. 7: Prototype of proposed Synchronous Buck Converter based charger



Fig. 8: Experimental setup of the battery charging system

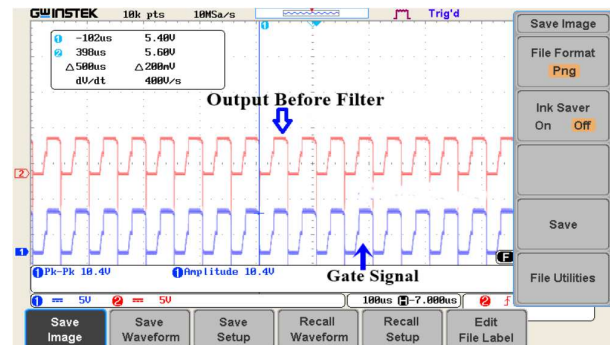


Fig. 9: Waveform for 48V output load connected with ASBC

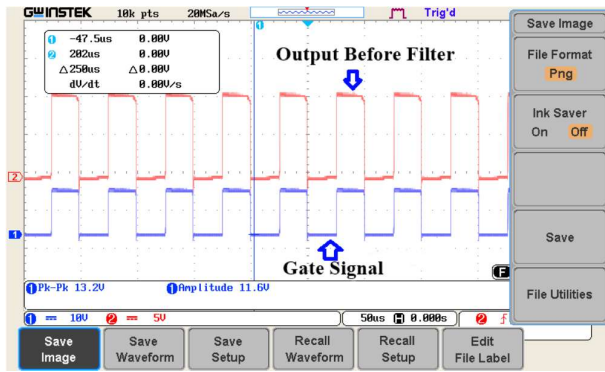


Fig. 10: Waveform for 48V output load connected with SBC

B. Gate Drive Pulses

Figures 9 and 10 shows the oscilloscope outcome for ASBC and SBC charging system, respectively. For both cases, the red signal indicates the output voltage waveform and the blue signal act as the gate signal. 48V.

V. PERFORMANCE COMPARISON

Table IV and Table V shows the experimental data obtained from the proposed Synchronous Buck Converter and Asynchronous Buck Converter charging system.

TABLE IV. Efficiency analysis for the SBC charging method

V_i (V)	I_i (A)	P_i (W)	V_o (V)	I_o (A)	P_o (W)	η (%)
103.9	1.12	116.368	49.4	1	49.4	42.451
104.5	1.45	151.525	49.8	2	99.6	65.731
105.8	1.70	179.586	50.4	3	151.2	84.065
106.9	2.00	213.800	50.9	4	203.6	95.229
107.9	2.50	269.750	51.4	5	257.0	95.273

TABLE V. Efficiency analysis for the ASBC charging method

V_i (V)	I_i (A)	P_i (W)	V_o (V)	I_o (A)	P_o (W)	η (%)
72.7	1.05	76.335	49.8	1	49.8	65.238
74.3	2.12	157.516	50.4	2	100.8	63.993
75.2	2.55	191.760	50.9	3	152.7	79.630
75.8	3.12	236.496	51.5	4	206.0	87.105
77.7	3.62	281.274	52.3	5	261.5	92.969

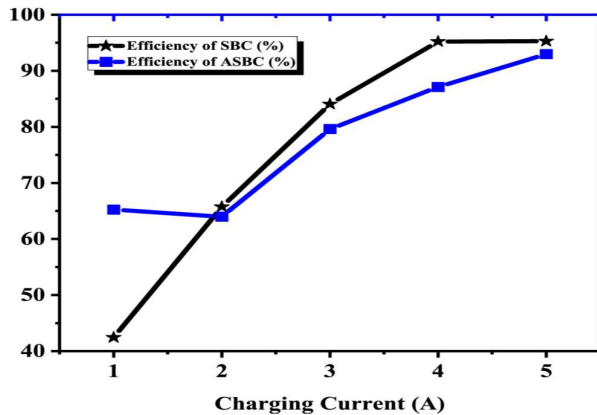


Fig. 11: Comparison of efficiency against charging current for proposed ASBC and SBC battery charging systems

The performance is analyzed based on the data taken from hardware prototype of Synchronous Buck Converter and Asynchronous Buck Converter. Figure 11 shows a graphical representation of system efficiency against charging current for the proposed SBC and ASBC battery charging system. It is found from the laboratory experiment that, in case of low

charging current rate (1A) the Asynchronous Buck Converter (ASBC) provides a better performance than the Synchronous Buck Converter (SBC). However, in the case of the high charging current rate, Synchronous Buck Converter (SBC) provides the best performance.

VI. CONCLUSION

In this paper, a detailed analysis of Solar-powered Asynchronous Buck Converter (ASBC) and Synchronous Buck Converter (SBC) for EVs battery charging applications is presented. A direct comparison of these two proposed charging techniques for EVs battery charging is evaluated for the same charging current. The simulated results and experimental findings suggest that the proposed Synchronous Buck Converter battery charging system produces better performance than the Asynchronous Buck Converter battery charging system. Additionally, these proposed systems show the least variation between simulation and experimental results, which established the sustainable design of charging system.

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